SYNTAX-AWARE SILICON: ARCHITECTURE SPECIFICATION v5.0

Document Class: Cryptographic Timestamp / Core IP Protection

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1. 3D Monolithic Multi-Valued Logic (MVL) Spatial Stacking

Legacy semiconductor design relies on flat 2D planar binary routing, causing massive propagation delays. LDH v5.0 introduces a **3D Monolithic Burger-Type Topography (3D-MBTT)** layered architecture structured symmetrically around a zero-potential balance reference node.

- **The Neutral Core Axis (L_0):** A central physical ground substrate (0.0V) handling real-time context-free grammar parsing, localized execution caching, and core hardware punctuation routines.
- **The Positive Divergence Stack (L_{+n}):** Discrete, vertically isolated processing layers operating at incremental positive voltage steps **($+V_n$)**. These layers are mapped exclusively to high-activation semantic processing clusters.
- **The Negative Divergence Stack (L_{-n}):** Symmetrical lower-level processing layers operating at decremental negative differential buffer zones **($-V_n$)**, acting as dynamic syntactic filters and state modulators.

2. Vertical Asynchronous Matrix Interconnects (VAMI)

Instead of standard planar serialization, the incoming symbolic string triggers localized vertical execution. Vertical micro-conductors intersect the L divergence stacks. Upon vector ingestion, specific layers resonate based on their predefined quantized voltage signatures, causing a vertical cross-layer current injection. The intersecting currents collapse directly onto the output bus as a unified, non-binary, complex analog **Syllabic Waveform**, processing high-entropy multi-state data packets in a **single deterministic clock cycle**.
